

STEP 1: ANALYSIS

1. Problem Restatement

The goal is to create a safety system that works automatically opens and closes the gates at a railway crossing based on logic. The system needs to lower the gates when a train is coming or when a car is still on the tracks. It should only lift the gates when it is fully safe. The concept is to keep traffic moving smoothly while making it as safe as possible.

2. Inputs and Outputs

Inputs:

- Train sensor (detects train approaching or passing)
- Vehicle sensor (detects if a vehicle is on the tracks)
- Track-clear sensor (confirms no vehicles remain)

Outputs:

- Gate status (lowered or raised)
- Warning signals (lights or alarms)

3. Context, Constraints, Stakeholders

Context: There is a considerable risk of accidents at railway crossings. Introducing a reliable logic system can help prevent possible collisions between trains and other vehicles.

Constraints:

1. **Technical:** Sensors must be accurate, gates must respond quickly.
2. **Economic:** Affordable implementation for many crossings.
3. **Social:** Must protect drivers, passengers, and pedestrians.
4. **Environmental:** Energy-efficient and weather-resilient
5. **Legal:** Must comply with road and railway safety regulations.

Stakeholders: Railway authorities, train operators, road users (drivers, cyclists, pedestrians), local government, and emergency services.